

♥ UNIT – 1 Answers (Clean + Exam-Ready)

1. Forces behind the evolution of Data Communication & Networking

- Growth of the Internet
 - Need for faster communication
 - Advances in hardware (fiber optics, wireless)
 - New applications: e-commerce, cloud, multimedia
 - Globalization + demand for real-time services
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2. Enterprise Applications

- Email, VoIP, Video conferencing
 - Cloud computing
 - File sharing
 - ERP, CRM systems
 - Remote access (VPN)
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3. Technologies Used

- Wired: Ethernet, Fiber
 - Wireless: Wi-Fi, Bluetooth
 - WAN: MPLS, Leased lines
 - Internet protocols: TCP/IP
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4. Five Components of Data Communication

1. **Message** – data
 2. **Sender** – device sending
 3. **Receiver** – device receiving
 4. **Medium** – cable, wireless
 5. **Protocol** – rules
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5. Types of Connections (Modes)

Simplex

One-way only (e.g., keyboard → monitor)

Half Duplex

Both ways but not at same time (Walkie-talkie)

Full Duplex

Both ways simultaneously (Phone call)

6. Types of Connections (Points)

- **Point-to-Point:** Single dedicated link
 - **Multipoint:** Multiple devices share one link
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7. Communication Methods

- **Unicast:** One-to-one
 - **Broadcast:** One-to-all
 - **Multicast:** One-to-group
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8. Network Topologies

- Bus
 - Ring
 - Star
 - Mesh
 - Hybrid
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9. Protocol — Need & Elements

Need:

Communication standardization, error control, flow control, compatibility

Elements:

- Syntax
 - Semantics
 - Timing
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10. TCP/IP Model (Layers + Functions)

1. **Application:** HTTP, FTP, DNS
 2. **Transport:** TCP/UDP, reliability
 3. **Network:** IP addressing & routing
 4. **Data Link:** Framing, MAC address
 5. **Physical:** Signals, media
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11. OSI Model (7 Layers)

1. Physical
 2. Data Link
 3. Network
 4. Transport
 5. Session
 6. Presentation
 7. Application
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12. Data & Signals

Digital signal: discrete values

Analog signal: continuous wave

13. Digital Signal Numericals

- **Bit rate = baud rate × bits/level**
 - **Bit length = propagation speed × bit duration**
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14. Transmission Impairments

- **Attenuation:** weakening (formula: $\text{dB} = 10 \log(P_2/P_1)$)
 - **Distortion:** change in signal shape
 - **Noise:** unwanted energy
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15. Line Coding

Unipolar: single voltage

Polar:

- RZ
- NRZ-L
- NRZ-I

Biphase: Manchester, Differential Manchester

16. Analog-to-Digital Encoding

PAM: samples amplitude

PCM: sampling → quantization → encoding

♥ UNIT – 2 Answers (Clean + Exam-Ready)

1. Bandwidth

Definition:

Maximum amount of data that can be transmitted per second.

Factors:

- Transmission medium
- Noise
- Distance
- Equipment quality

Formula:

Bit rate = $2 \times \text{bandwidth} \times \log_2(L)$

2. Throughput

Definition:

Actual data transfer rate achieved.

Factors:

- Network congestion
 - Errors
 - Protocol overhead
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3. Delay (Latency)

Total delay = **dproc + dqueue + dtrans + dprop**

Components:

- **Processing delay (dproc):** time to check header
 - **Queuing delay (dqueue):** waiting in buffer
 - **Transmission delay (dtrans):** L / R
 - **Propagation delay (dprop):** distance / speed
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4. Bandwidth Efficiency

Formula:

Efficiency = (data rate / bandwidth)

Methods to improve:

- Multiplexing (FDM, TDM, SDM)
 - Compression
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5. Bit Rate vs Baud Rate

- **Bit rate:** bits per second
- **Baud rate:** signal changes per second

Formula:

Bit rate = baud rate × bits per signal

6. Multiplexing

Definition:

Allowing multiple signals to share one medium.

Types:

- **FDM:** frequency channels
 - **TDM:** time slots
 - **SDM:** statistical time allocation
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7. DLL in OSI

Sublayers:

- **LLC:** logical link control
- **MAC:** media access control

Functions:

Framing, Addressing, Error control, Flow control, Access control

8. Link Layer Addressing

- **Unicast:** one device
 - **Multicast:** group
 - **Broadcast:** all devices
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9. ARP

Working:

- Sender broadcasts ARP request
 - Target replies with MAC address
 - Sender maps IP → MAC
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10. Standard Ethernet

Characteristics:

- Frame format
- Preamble + SFD
- DA + SA
- Length/Type
- Data
- CRC

Addresses:

Unicast, Multicast, Broadcast

11. MAC Methods

- **CSMA:** carrier sensing
 - **CSMA/CD:** collision detection (wired)
 - **CSMA/CA:** collision avoidance (wireless)
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12. WLAN

Types:

- **BSS:** Independent / Infrastructure
- **ESS:** multiple BSS

Features:

Mobility, wireless access, multiple addresses

13. Framing Techniques

- Character count
 - Byte stuffing, Bit stuffing
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♥ UNIT – 3 Answers (INTERNetworking + IP + Routing)

1. INTERNetworking – Introduction

Internetworking = connecting multiple networks so they act like one big network.

2. Principles of Internetworking (9 Principles)

1. Interconnect networks
 2. Provide universal addressing
 3. Support best-effort delivery
 4. Routing support
 5. Hide network differences
 6. Scalability
 7. Reliability
 8. Efficiency
 9. Layered architecture
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3. Network Layer

Introduction:

Third OSI layer, manages packet forwarding.

Features:

- Logical addressing
- Routing
- Packetization

Services:

- Connectionless service
- Best-effort delivery

Devices:

Router, Layer-3 Switch

4. Internet Protocol (IPv4)

Basics:

Connectionless, unreliable, best-effort.

IPv4 Classes:

- A, B, C: unicast
- D: multicast
- E: experimental

Datagram Header:

Version, IHL, Type of Service, Total Length, ID, Flags, Fragment Offset, TTL, Protocol, Header Checksum, Source IP, Destination IP.

5. Public vs Private IP

Private IP ranges:

- 10.0.0.0 – 10.255.255.255
- 172.16.0.0 – 172.31.255.255
- 192.168.0.0 – 192.168.255.255

Public = routable on internet.

Private = used inside LAN.

6. Subnet Mask Numericals

Network Address:

AND (IP, Mask)

Broadcast Address:

Set all host bits to 1.

No. of Hosts:

$2^n - 2$ (n = host bits)

7. CIDR (Classless Inter-Domain Routing)

What:

Classless addressing using prefix (/n)

Advantages:

- Removes class waste
- Better routing
- Aggregation (supernetting)

CIDR → Subnet Mask Conversion:

/24 → 255.255.255.0

/26 → 255.255.255.192 etc.

CIDR Numericals:

- Hosts = $2^{(32 - \text{prefix})} - 2$
 - Network range = first to last usable
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8. VPN (Virtual Private Network)

What:

Secure tunnel over public internet.

Why:

Privacy, remote access, secure communication.

How it Works:

Data is encrypted → sent through tunnel → decrypted at destination.

Uses:

Remote working, secure browsing, intranet access.

9. IPSec

Why:

Security for IP packets.

Components:

- **AH** (Authentication Header)
- **ESP** (Encapsulating Security Payload)

How it Works:

Encrypt + authenticate packets at network layer.

Advantages:

Strong security, supports VPN.

10. DHCP (Dynamic Host Configuration Protocol)**Why:**

Auto-assign IPs → no manual work.

Process:

1. **DHCPDISCOVER**
2. **DHCPOFFER**
3. **DHCPREQUEST**
4. **DHCPACK**

Packet fields:

Op, HTYPE, HLEN, HOPS, XID, FLAGS, CIADDR, YIADDR, SIADDR, GIADDR, CHADDR, options.

11. RIP (Routing Information Protocol)**What:**

Distance Vector protocol.

Characteristics:

- Uses hop count
- Max 15 hops
- Slow convergence

Types:

- RIPv1
- RIPv2

How it Works:

Routers broadcast full routing table every 30 secs.

12. OSPF (Open Shortest Path First)

What:

Link State protocol.

Features:

- Uses Dijkstra algorithm
- Fast convergence
- Supports areas

Areas:

Backbone Area 0 + other areas

How it Works:

Routers exchange LSAs → build topology → compute shortest path.

Advantages:

Scalable, supports VLSM, more efficient than RIP.

13. OSPF vs RIP

Feature	RIP	OSPF
Type	Distance Vector	Link State
Metric	Hop count	Cost
Convergence	Slow	Fast
Scalability	Low	High
Algorithm	Bellman-Ford	Dijkstra

♥ UNIT – 4 Answers (Transport Layer, DNS, HTTP, Compression, VoIP, SIP)

1. Transport Layer – Introduction

Provides end-to-end communication between processes.

Functions: segmentation, reassembly, flow control, error control.

2. TCP (Transmission Control Protocol)

Features:

- Reliable
- Connection-oriented
- Flow + congestion control

Header Includes:

Source port, Dest port, Seq no., Ack no., Offset, Reserved, Flags, Window, Checksum, Urgent Pointer.

3. UDP (User Datagram Protocol)

Features:

- Unreliable
- Connectionless
- Faster, lightweight

Header:

Source port, destination port, length, checksum.

4. DNS (Domain Name System)

What:

Converts domain names → IP addresses.

Types:

- Generic (.com, .org)
- Country (.in, .us)

- Reverse mapping (IP → name)
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5. HTTP

Request + Response Format:

Both have **start line + header + blank line + body**

Methods:

GET, POST, PUT, DELETE, HEAD, OPTIONS

Status Codes:

200 OK, 301 Moved, 404 Not Found, 500 Server Error

6. Compression

Lossy:

Losing some data → smaller size (jpg, mp3)

Pros: high compression

Cons: reduced quality

Lossless:

No data loss (png, zip)

Pros: perfect recovery

Cons: bigger size

7. MPEG Audio Compression – Steps

1. Filter bank
 2. Psychoacoustic model
 3. Bit allocation
 4. Coding (Huffman)
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8. MPEG Video Compression – Steps

1. Frame types (I, P, B)
2. Motion estimation

3. DCT transform
4. Quantization
5. Entropy coding

9. SIP (Session Initiation Protocol)

Features:

Text-based, similar to HTTP.

Working:

Invite → Ringing → OK → ACK → Session → BYE

Components:

User Agent, Proxy Server, Registrar

Advantages:

Simple, flexible, scalable

Disadvantages:

Less security, depends on other protocols for media

10. VoIP

Features:

Voice over internet, cheap, flexible.

Working:

Voice → Encode → Packetize → Transmit → Decode

Advantages:

Low cost, mobility, integration.

Disadvantages:

Depends on internet, latency, jitter issues